

Past Papers

Short Questions

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I- Partition Chromatography

Type of chromatography in which stationary phase is liquid and mobile phase is also liquid is called

P.C. When solvent is liquid, it is called L-L chromatography. When solvent is gas it is called G-L chromatography.

Absorption Chromatography

Type of chromatography in which stationary phase is solid and mobile phase is liquid/gas is called

A.C. When moving phase is liquid, it is called liquid-solid chromatography. When it is gas it is called G-S chromatography.

2- Binders in TLC

A variety of TLC adsorbents are easily available. Silica is used for separation of classes of compounds. Binding agents (Binders) i.e. Plaster of Paris are incorporated with silica used to hold adsorbent on paper.

3- Limitations of TLC

- (i) Lack of reproducibility of R_f values
- (ii) It deals with small amount of material. It can't be used at large scale / industrial scale

4- Characteristics of adsorbents in column chromatography

- Alumina and magnesia have been found to give satisfactory separation
- Other adsorbents like Na_2SO_4

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MgSO₄, Sucrose, Starch and Charcoal are used in specific Separation.

- Starch and Sucrose are used in separation of chlorophylls.

5- Faster Technique [TLC, PC]

Although TLC is very similar to PC but it is considered to be superior because it is faster technique.

In TLC separation is sharp. It takes less time (15-50 min) and small amount of substances (0.4 mg).

6- R_f value

Retention / Retention factor

$$R_f = \frac{\text{Distance travelled by solute}}{\text{Distance travelled by solvent}}$$

Its value is less than one.
It has no unit.

7- Normal Phase Chromatography

It uses polar stationary phase and non-polar mobile phase. It can't be reused.

Eluents are expensive. Effective for separation of structural isomers.

Reverse Phase Chromatography

It uses non-polar stationary phase and polar mobile phase. It can be reused.

Eluents are inexpensive. Effective for separation of homologs.

8- Beer-Lambert's Law

The rate of decrease of intensity of monochromatic beam of light w.r.t thickness of medium is directly proportional to intensity of light and conc of absorbing substance in

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Solution.

$$\log_{10} \frac{I_0}{I} \propto c \cdot l$$

9- Filters

The filters are pieces of glass which transmit limited λ ranges of spectrum.

It operates by absorbing restricted band of radiation from continuous sources.

Monochromator

A monochromator consists of entrance and exit slits, a prism or grating and mirrors to deflect or focus radiation.

The prism or grating separates polychromatic radiation into monochromatic radiation.

The prism is made up of

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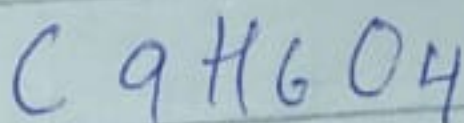
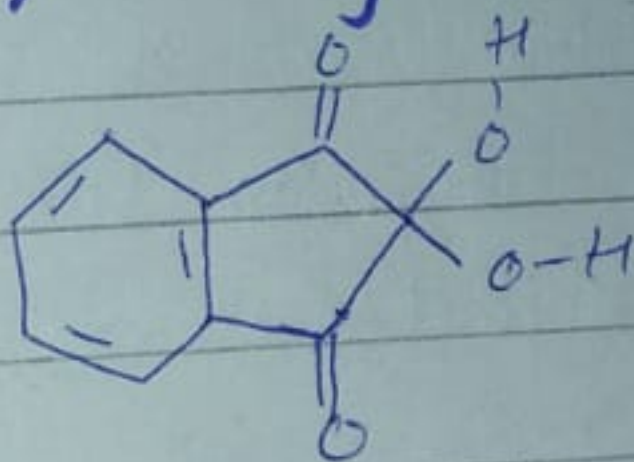
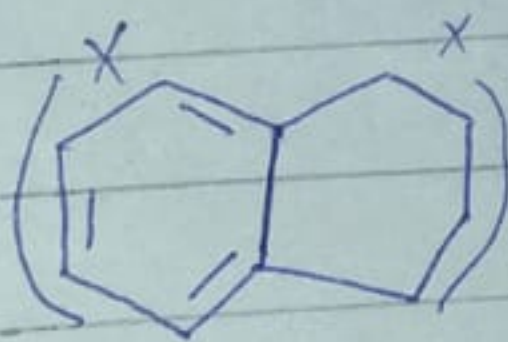
glass and fused quartz. Grating
is made from soft metal
like "Al."

10 - Charge Transfer absorption

When an e^- is transferred
from atom or group to another
atom or group in molecule,
the process is called charge
transfer absorption.

e.g. In MnO_4^- , e^- is transferred
from non-bonding orbital to
of d orbital of Mn.

11 - Formula of ninhydrin



12- Spectrometer

A spectrometer is a device for measuring wavelengths of light over a wide range of electromagnetic spectrum. It is widely used for spectroscopic analysis of sample materials.

Spectrophotometer

The commonly used instrument for study of absorption in IR its region is called spectrophotometer.

13- Precision

- Closeness of two measured values with same instrument
- There is no relation of precision with accuracy

Accuracy

- Closeness of measured value with true value
- It is strongly related with precision but not always

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It is validate instrument.

It is validate the instrument as well as sample.

14) Standard Deviation

Standard deviation is square root of sum of square of deviation divided by one less than the number of pieces of data in set.

$$S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N-1}}$$

x_i = individual measurement

$\bar{x}$ = mean value of all measurements

N = num of measurements.

15-Confidence Limit

Either of two numbers that specify the end point of confidence interval.

16- Relative Standard Deviation

It is ratio of Standard deviation of mean value to Average value

$$R.S.D = \frac{S}{\bar{x}} \times 100$$

Absolute error

The error which gives absolute difference of experimental result from true result is called absolute error

$$E_a = x_2 - x_t$$

x_2 = Individual result

x_t = True result

E_a = Absolute error

17- Molality

It is number of moles of solute dissolved

Molality

It is num of moles per kg of solvent

in one liter
of solution

• It is represented
by M

• $M = \frac{\text{moles of solute}}{\text{Liter of solution}}$

• Its unit is
 moles/L

• It is represented
by m

• $m = \frac{\text{moles of solute}}{\text{mass of solvent in kg}}$

• Its unit is
 mole/kg

18. Relative Error

The error which gives the relative difference of experimental result from true result

$$E_x = \frac{x_i - x_t}{x_t} \times 100$$

Relative Accuracy

Relative closeness of measured value with true value.

19- Median:

For a set of data, the middle value in any order is called median.

10.1, 10.2, 10.3, 10.5, 10.8
 ↓
 mean of two
 middle values

Mean:

The average of data is called mean

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

20- Molecular Spectroscopy

M.S is the study of interaction of electromagnetic radiation with matter, impart information regarding molecular structure and chemical properties. Matter can be atoms molecules, atomic

or molecular ions or solids.

21- Wavelength Range

UV $\rightarrow 10^{-5}$

Vislight $\rightarrow 10^{-4}$

22- Advantage of double beam Spectrophotometer

A single-beam spectrophotometer is inconvenient because the sample and reference must be placed alternatively in beam for measurement at multiple wavelengths. The reference must be run at each wavelength so it is poorly suited to measure absorbance as function of time as in kinetics experiment because both source intensity and detector response slowly drift. In double-beam, light is alternatively passes through

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the sample of reference directly by rotating mirror.

23 One tailed Test

One tailed is used either to establish whether one experimental value is significantly greater than other or other way around

Two tailed Test

Two tailed test is used to establish whether there is significant diff b/w two values being compared whether one is higher or lower than other being not specified

24- Rf values

Rf values are calculated immediately after spots are visualized so that effect of

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temp and pressure may destroy
value

25- Euotopic series

An euotopic series is listing of various compounds in order of eluting power for given adsorbent. This often happens when element adsorb onto stationary phase displacing Analyte.

26- Preparation of ppm solution of KMnO_4

ppm solution of KMnO_4 is prepared by dissolving 1mg of KMnO_4 in 1L of distilled water.

27- Visualization

Visualization is basically appearance of any spot. In chromatography the term visualization is used

to see spots on chromatogram.
There are many visualizing agents used to visualize different spots.

28) Paper's ability to hold water

Paper is network made from randomly deposited cellulose fibers which are about 1-3 mm long and 20-30 μm in diameter.

The fibers are hollow tube-like structures. The fiber wall consists of cellulose, hemicellulose and lignin. Although most of lignin is removed before usage we can take into account water as fiber so water can exist in form of fiber wall.

29) Control Chart

A control chart is time

plot of measured quantity that is assumed to be constant for purpose of ascertaining that measurement remains within statistically acceptable range.

e.g. $\bar{X}$ -Bar R -Bar

30) Kieselguhr

A form of diatomaceous earth used in various manufacturing and laboratory processes chiefly as a filter, filler or insulator.

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$$C.L = \bar{x} \pm t_s \frac{s}{\sqrt{N}}$$

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